

Lesson 7 - FCFS vs SJF vs Round Robin

Question 1 - Arrival time Zero

Process	Burst Time
P1	5
P2	3
P3	4

1. FCFS



- Completion time = 12
- Turnaround time
 1. $p1 = 5 - 0 = 5$
 2. $p2 = 8 - 0 = 8$
 3. $p3 = 12 - 0 = 12$
- Wait time
 1. $p1 = 5 - 5 = 0$
 2. $p2 = 8 - 3 = 5$
 3. $p3 = 12 - 4 = 8$
- Average wait time = $13/3 = 4.33$

2. SJF



- Execution order = p2,p3,p1
- completion time = 12
- Turnaround time
 1. $p1 = 12 - 0 = 12$
 2. $p2 = 3 - 0 = 3$
 3. $p3 = 7 - 0 = 7$
- Wait time
 1. $p1 = 7 - 0 = 7$
 2. $p2 = 0 - 0 = 0$
 3. $p3 = 3 - 0 = 3$
- Average wait time = $10/3 = 3.33$

3. Round Robin



- Turnaround time
 1. $p1 = 11 - 0 = 11$
 2. $p2 = 6 - 0 = 6$
 3. $p3 = 12 - 0 = 12$
- Wait time
 1. $p1 = 11 - 5 = 6$

$$2. p2 = 6 - 3 = 3$$

$$3. p3 = 12 - 4 = 8$$

- Average wait time = $17/3 = 5.67$

Comparison

	FCFS	SJF	Round Robin
Average wait time	4.33	3.33	5.67

- *SJF* provides the *lowest average wait time* while *Round Robin* has the *longest wait time*.

Question 2 - With Arrival time

Process	AT	BT
P1	0	5
P2	1	3
P3	2	4

1. FCFS



- Turnaround time
 1. $p1 = 5 - 0 = 5$
 2. $p2 = 8 - 1 = 7$
 3. $p3 = 12 - 2 = 10$

- Wait time
 1. $p1 = 5 - 5 = 0$
 2. $p2 = 7 - 3 = 4$
 3. $p3 = 10 - 4 = 6$
- Average wait time = $10/3 = 3.33$

2. SJF

SJF non - preempt



- Turnaround time
 1. $p1 = 5 - 0 = 5$
 2. $p2 = 8 - 1 = 7$
 3. $p3 = 12 - 2 = 10$
- Wait time
 1. $p1 = 5 - 5 = 0$
 2. $p2 = 7 - 3 = 4$
 3. $p3 = 10 - 4 = 6$
- Average wait time = $10/3 = 3.33$

SJF Preempt



- Turnaround time

1. $p1 = 8 - 0 = 8$
 2. $p2 = 4 - 1 = 3$
 3. $p3 = 12 - 2 = 10$
- Wait time
 1. $p1 = 8 - 5 = 3$
 2. $p2 = 3 - 3 = 0$
 3. $p3 = 10 - 4 = 6$
 - Average wait time = $9/3 = 3$

3. Round Robin



- Turnaround time
 1. $p1 = 11 - 0 = 11$
 2. $p2 = 6 - 1 = 5$
 3. $p3 = 12 - 2 = 10$
- Wait time
 1. $p1 = 11 - 5 = 6$
 2. $p2 = 5 - 3 = 2$
 3. $p3 = 10 - 4 = 6$
- Average wait time = $14/3 = 4.66$

Comparison

	FCFS	SJF non-preempt	SJF Preempt	Round Robin
Average wait time	3.33	3.33	3	4.66

- *SJF preempt* provides the *lowest average wait time* while *Round Robin* has the *longest wait time*.